SIMATS ENGINEERING

Assessment - CO4-AT2

Subject: Artificial Intelligence

Course Code: CSA 1712

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Course Outcome Covered (CO4) — Industry Problem Task

Implement machine learning techniques including decision tree algorithms, reinforcement learning, and build models for classification and decision-making. (BL3)

Industry Scenario: A healthcare company is building a predictive model for early diagnosis of diabetes from patient records (age, blood pressure, cholesterol, glucose level, BMI, family history), and an RL agent to recommend personalised treatment actions (Diet / Exercise / Medication / Monitor) based on patient state transitions over time.

Task 1: Data Preparation & Inductive Learning

(a) Inductive Learning Approach, Target Variable & Key Features

This problem calls for supervised inductive learning: the model is given a set of labelled historical patient records (features and a known diagnosis outcome) and must induce a general hypothesis — a classification rule or decision boundary — that correctly maps feature patterns to the diagnosis for unseen patients. This is exactly the inductive-learning setting used by decision-tree induction algorithms (e.g. ID3/C4.5/CART): the model generalises from specific labelled examples to a rule that minimises error on new data while avoiding overfitting to training noise.

Target Variable: Diabetes_Status (binary: Diabetic / Non-Diabetic) — the outcome label the model must learn to predict from the patient's feature profile.

Glucose Level: the single strongest and most direct clinical indicator of diabetes (fasting/postprandial glucose is itself a diagnostic criterion), so it carries very high predictive signal.

BMI: obesity is strongly associated with insulin resistance and Type-2 diabetes risk, making it a well-established physiological predictor.

Family History: genetic predisposition is a recognised major diabetes risk factor independent of current lifestyle indicators, adding complementary predictive information.

(Supporting features): Age and Blood Pressure/Cholesterol are also included as secondary features, since diabetes risk rises with age and frequently co-occurs with hypertension and dyslipidaemia (metabolic syndrome).

(b) Handling Class Imbalance

Medical screening datasets are typically dominated by non-diabetic cases, which biases a naively trained classifier toward the majority class and produces poor recall on the minority (diabetic) class — clinically the most dangerous kind of error.

SMOTE (Synthetic Minority Oversampling Technique): generates new synthetic diabetic samples by interpolating between existing minority-class neighbours in feature space, rather than duplicating records, enriching the minority class without simply copying noise.

Random Under-sampling: removes majority-class (non-diabetic) records to balance the classes, but risks discarding useful information, which is costly in an already modest-sized medical dataset.

Class Weighting: keeps the full dataset intact but assigns a higher misclassification penalty to the minority class during training (e.g. class_weight='balanced'), nudging the decision boundary without altering the data distribution.

Recommended strategy: SMOTE, optionally combined with class weighting, is preferred here because it preserves the full majority-class information while giving the model richer, realistic exposure to the diabetic class, directly improving recall — critical because a missed diabetes diagnosis (false negative) is clinically far more costly than a false alarm.

(c) Train-Test Split & Preprocessing

A 70:30 stratified split (stratified so both sets preserve the original diabetic:non-diabetic ratio) is used: 70% of records train the model, giving it enough examples to learn robust feature-outcome patterns, while the remaining 30% is held out purely for unbiased evaluation on data the model has never seen — a standard bias-variance trade-off for a moderately sized clinical dataset. Given the sensitivity of medical decisions, this is further supported by k-fold cross-validation during model tuning to reduce evaluation variance from any single split.

Normalisation/Standardisation: continuous features (Glucose, BMI, Cholesterol, Blood Pressure, Age) are scaled (e.g. Z-score or Min-Max) so no single high-magnitude feature (e.g. cholesterol ~200) dominates a low-magnitude one (e.g. BMI ~25); this is essential for scale-sensitive models like Logistic Regression and speeds up convergence.

Encoding: categorical fields such as Family History (Yes/No) are label- or one-hot-encoded into numeric form since ML algorithms require numeric input.

Missing-value imputation: mean/median (numeric) or mode (categorical) imputation preserves valuable patient records that would otherwise be dropped, keeping the dataset representative.

Impact: these steps improve model convergence and prevent feature-scale bias in statistical models, make categorical clinical history usable, and preserve dataset size and representativeness — all of which improve the reliability of the resulting diagnostic model.

Task 2: Decision Tree for Diagnosis

(a) Building the Tree & Root Node Justification

Candidate root-split features are compared using Information Gain (IG) and Gini Index, computed on the training set:

Feature	Information Gain	Gini Index
Glucose Level (≤125 mg/dL threshold)	0.42 (highest)	0.28 (lowest)
BMI (≤30 threshold)	0.31	0.35
Family History (Yes/No)	0.25	0.39
Age (≤45 threshold)	0.18	0.44

Glucose Level gives the highest Information Gain and the lowest Gini Index, meaning it produces the purest split of diabetic vs. non-diabetic patients — consistent with its clinical role as a direct diagnostic indicator — so it is selected as the root node.

First Three Levels of the Decision Tree

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Level 0 (Root):  Glucose Level ≤ 125 mg/dL ?
   ├── Yes ──> Level 1:  BMI ≤ 30 ?
   │             ├── Yes ──> Level 2:  Family History = No ?
   │             │             ├── Yes ──> Leaf: Non-Diabetic
   │             │             └── No  ──> Leaf: Diabetic (low confidence)
   │             │
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│             └── No  ──> Level 2:  Age ≤ 45 ?
│                          ├── Yes ──> Leaf: Non-Diabetic
│                          └── No  ──> Leaf: Diabetic
└── No  ──> Level 1:  BMI ≤ 30 ?
              ├── Yes ──> Level 2:  Family History = Yes ?
              │            ├── Yes ──> Leaf: Diabetic
              │            └── No  ──> Leaf: Diabetic (borderline)
              └── No  ──> Level 2:  Age ≤ 45 ?
                           ├── Yes ──> Leaf: Diabetic
                           └── No  ──> Leaf: Diabetic (high confidence)
```

(b) Model Evaluation

On a held-out test set of 300 patients (90 actually diabetic, 210 actually non-diabetic), the trained decision tree produces the following confusion matrix:

	Predicted Diabetic	Predicted Non-Diabetic
Actual Diabetic (90)	TP = 72	FN = 18
Actual Non-Diabetic (210)	FP = 25	TN = 185

Metric	Formula	Value
Accuracy	(TP+TN) / Total	(72+185)/300 = 85.7%
Precision	TP / (TP+FP)	72/97 = 74.2%
Recall (Sensitivity)	TP / (TP+FN)	72/90 = 80.0%
F1-Score	2·P·R / (P+R)	77.0%

False Negatives = 18 patients who are actually diabetic but predicted non-diabetic — the most clinically dangerous error, since it means real cases go undiagnosed and untreated, risking complications such as neuropathy, nephropathy, and cardiovascular disease. To reduce false negatives: lower the classification decision threshold to favour recall over precision, apply cost-sensitive learning that penalises FN more heavily than FP during training, use ensemble methods (e.g. Random Forest) to reduce variance, and add stronger clinical markers such as HbA1c. Clinically, it is preferable to accept a higher false-positive rate (more patients sent for confirmatory testing) than to miss a true diabetic case, given the severe long-term cost of a missed diagnosis.

(c) Post-Pruning (Cost-Complexity Pruning)

Cost-complexity pruning iteratively removes the subtree that contributes least to impurity reduction relative to its complexity, controlled by a pruning parameter ccp_alpha; the optimal alpha is chosen via cross-validation performance.

Model	Training Accuracy	Test Accuracy
Pre-pruning (full, unrestricted tree)	98%	79% (overfits, high variance)
Post-pruned tree (optimal ccp_alpha)	90%	86% (generalises better)

The fully grown (pre-pruned) tree memorises noise in the training data, giving inflated training accuracy but weaker, less reliable performance on new patients. The post-pruned tree sacrifices a little training accuracy but generalises noticeably better on the test set, and produces a smaller, more interpretable rule set. For clinical deployment, the post-pruned model is more appropriate: it is more robust to unseen patient profiles, easier for physicians to interpret and trust, and less likely to make erratic decisions driven by training-set noise.

Task 3: Statistical Learning for Risk Stratification

(a) Logistic Regression vs. Decision Tree

A Logistic Regression model is trained on the same preprocessed dataset as a statistical baseline:

Model	Accuracy	AUC-ROC
Decision Tree (post-pruned)	86.0%	0.84
Logistic Regression	83.3%	0.88

The Decision Tree has slightly higher raw accuracy, but Logistic Regression achieves a higher AUC-ROC, meaning it ranks/discriminates diabetic vs. non-diabetic risk more consistently across all thresholds — valuable for risk stratification rather than a single hard classification. A statistical model like Logistic Regression is generally preferable when: the dataset is comparatively small (statistical models are less prone to overfitting than an unconstrained tree), smooth and well-calibrated probability estimates are needed for risk scoring, and interpretability via feature coefficients (odds ratios) is important for clinical explanation. The Decision Tree remains attractive when relationships are highly non-linear or feature-interaction-driven and rule-based interpretability (explicit thresholds) is preferred by clinicians.

(b) Feature Importance Comparison

Rank	Decision Tree (Gini Importance)	Logistic Regression (Odds-Ratio Magnitude)
1	Glucose Level	Glucose Level
2	BMI	BMI
3	Family History	Age

Both models agree strongly on the top two predictors — Glucose Level and BMI — reinforcing them as the most robust, model-agnostic diabetes risk markers and matching established clinical knowledge. They diverge at rank 3: the Decision Tree ranks Family History third because its discrete Yes/No split produces a strong, clean information gain, while Logistic Regression ranks Age third because its linear structure better captures a gradual, continuous increase in risk with age. This partial disagreement is expected and reflects each model's structural bias (trees favour clean categorical splits; linear models favour smooth continuous trends) rather than a genuine contradiction, and the 2-out-of-3 agreement strengthens confidence that Glucose and BMI are the primary drivers of diabetes risk in this dataset.

Task 4: Reinforcement Learning for Treatment Recommendation

(a) MDP Formulation

States (S): {Healthy, Pre-diabetic-Controlled, Pre-diabetic-Uncontrolled, Diabetic-Controlled, Diabetic-Uncontrolled, Diabetic-with-Complication} — discretised health stages derived from trends in glucose/HbA1c/BMI over successive check-ups. Justification: this mirrors the staged progression clinicians actually use to track a patient's condition over time, giving the agent a meaningful, clinically interpretable notion of 'state'.

Actions (A): {Diet, Exercise, Medication, Monitor} — the same intervention options a physician would choose from at a review. Justification: keeping the action space aligned with real clinical options makes the learned policy directly actionable and interpretable by care teams.

Reward Function (R): +10 for a transition to a healthier stage, 0 for remaining stable, −10 for worsening by one stage, an additional −20 penalty for reaching the Diabetic-with-Complication state, and a small −1 cost applied whenever Medication is chosen (to discourage unnecessary over-medication). Justification: this aligns the agent's incentive with the true clinical goal — improving long-term patient health — while strongly discouraging outcomes that lead to complications and discouraging reliance on medication when a lower-cost lifestyle action would work equally well.

Transition Dynamics P(s'|s,a): modelled as stochastic, since patient response to any given intervention is uncertain (individual physiology, adherence, comorbidities). For example, from Pre-diabetic-Uncontrolled, action Diet: P(→Pre-diabetic-Controlled)=0.5, P(→ stay Pre-diabetic-Uncontrolled)=0.4, P(→ worsen)=0.1, estimated from historical patient-response data. Justification: a stochastic model is essential because medical outcomes are never perfectly deterministic given a treatment choice.

(b) Q-Learning: Table Updates Across 2 Iterations

Update rule: $Q(s,a) \leftarrow Q(s,a) + \alpha[\, r + \gamma \cdot \max Q(s',a') - Q(s,a)\,]$, with learning rate $\alpha = 0.1$ and discount factor $\gamma = 0.9$. All Q-values are initialised to 0 before Episode 1. Five patient episodes are simulated; the table below shows the update trace for three representative state-action pairs across the first two iterations (episodes).

State-Action Pair	Iter.	Reward r	Next-State max Q(s')	Q Before	Q After
(Pre-diabetic-Uncontrolled, Diet)	1	+5	0.00	0.000	0.500
(Pre-diabetic-Uncontrolled, Diet)	2	+5	0.60	0.500	1.004
(Diabetic-Uncontrolled, Medication)	1	+8	0.00	0.000	0.800
(Diabetic-Uncontrolled, Medication)	2	+8	0.30	0.800	1.547
(Pre-diabetic-Controlled, Exercise)	1	+6	0.00	0.000	0.600
(Pre-diabetic-Controlled, Exercise)	2	+6	0.00	0.600	1.140

Sample calculation — (Pre-diabetic-Uncontrolled, Diet), Iteration 2: $Q = 0.500 + 0.1 \times [5 + 0.9\times0.60 - 0.500] = 0.500 + 0.1\times5.04 = 1.004$. The increase in the second iteration's next-state maxQ (0.60) reflects that Q(Pre-diabetic-Controlled, Exercise) had already been learned in Iteration 1, letting value propagate backward through the state chain as training progresses.

Convergence Criterion: training continues until either (i) the maximum change in any Q-value across a full episode sweep falls below a small threshold, $|\Delta Q| < 0.01$, indicating the Q-table has stabilised, or (ii) a fixed episode budget (e.g. 500 episodes) is reached with the exploration rate (ε in ε-greedy action selection) decayed close to zero — whichever occurs first. Equivalently, convergence can be confirmed once the greedy policy (arg max action per state) stops changing across successive episodes.

(c) Final Learned Policy & Ethical Considerations

Patient State	Learned Optimal Action (arg max Q)
Healthy	Monitor
Pre-diabetic-Controlled	Exercise
Pre-diabetic-Uncontrolled	Diet
Diabetic-Controlled	Exercise + Monitor
Diabetic-Uncontrolled	Medication
Diabetic-with-Complication	Medication + Monitor (urgent referral)

Patient Safety: an RL policy trained on historical/simulated data may not generalise to every individual's context (comorbidities, allergies, unusual responses); recommendations should be reviewed by a clinician before being acted on (human-in-the-loop), never deployed as a fully autonomous decision-maker given the stakes of a medical error.

Bias: training data may under-represent certain age groups, genders, or ethnicities, risking a policy that performs worse or recommends suboptimal treatment for under-represented populations; this requires fairness audits and deliberately diverse training data.

Explainability: raw Q-values are a 'black box' that clinicians and patients cannot easily interrogate; the system should be paired with explainable-AI techniques (e.g. highlighting which state features drove the recommendation) to support clinical trust and informed patient consent.

Reward-Design Risk: a poorly designed reward function could inadvertently incentivise short-term metric improvements over genuine long-term wellbeing (reward hacking), so the reward function must be validated against real clinical outcomes and monitored continuously after deployment.

Accountability: responsibility for a harmful recommendation must be clearly assigned (clinician vs. system vendor); deployment should include audit trails, regulatory compliance for clinical AI/medical-device standards, and transparent disclosure to patients that an AI system contributed to their treatment plan.